Operational Definition of the Phoenix Sepsis Score

August 21, 2026

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1 Objective

Provide formal notation for the operational definition of the Phoenix Sepsis Score as published in Sanchez-Pinto et al. [2024] and Schlapbach et al. [2024] along with several exploratory aggregation schemes. The different scoring methods are based on when and how data is aggregated. The differences in the aggregation methods can dramatically change the Phoenix score and the published cut point of two or more points for sepsis may not be applicable to all aggregations.

2 Notation

This section establishes the notation used throughout the operational definition. It first lists common symbols and then defines the helper functions used to carry values forward, restrict values to look-back windows, and evaluate indicator functions with missing data.

2.1 Symbols

This section is a quick guide to the symbols used throughout the document. Detailed definitions for the functions will follow in § 2.2. See Table 1.

Table 1: Notation used in the operational definition of the Phoenix Sepsis Score.

Symbol	Meaning	Example or Notes
Sets and Membership		
$\{\cdot\}$	Set notation used to define collections of values satisfying a condition.	$\{x : x \leq t\}$ denotes the set of values x less than or equal to t. Also, $\boldsymbol{A} = \{1, 3, 5\}$ is read as "the set A consisting of the elements 1, 3, and 5."
$\emptyset$	Empty set.	Represents a set containing no elements.
$\in$	Set membership operator.	$t \in T$ indicates that time t lies within the time window T.
$\boldsymbol{A} \setminus \boldsymbol{B}$	Remove elements of a set	Example $\{1, 2, 3\} \setminus \{2\} = \{1, 3\}$
$\cup$	Union	$\{1, 2\} \cup \{2, 3\} = \{1, 2, 3\}$
Intervals		
(a, b)	Open interval.	$X \in (a, b) \equiv a < X < b$.
$[a, b)$	Left-closed, right-open interval.	$X \in [a, b) \equiv a \leq X < b$.
$(a, b]$	Left-open, right-closed interval.	$X \in (a, b] \equiv a < X \leq b$.
$[a, b]$	Closed interval.	$X \in [a, b] \equiv a \leq X \leq b$.
Definitions, Equality, and Logic		
$:=$	Definition operator indicating that the quantity on the left is defined by the expression on the right.	$f(x) := x^2$ defines the function $f(x)$ as x^2.
$=$	Equality relation.	$2 + 3 = 5$.

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Symbol	Meaning	Example or Notes
$\equiv$	Logical equivalence or identity.	$A \equiv B$ means that A and B are equivalent statements.
$\vee$	Logical OR operator.	$A \vee B$ is true if either A or B is true.
$\wedge$	Logical AND operator.	$A \wedge B$ is true if both A and B are true.
$\forall$	Universal quantifier ("for all").	$\forall t \in T$ means for every t in the time window T.
$\exists$	Existential quantifier ("there exists").	$\exists t \in T$ means for some time t in T.
Arithmetic and Functional Operators		
$\sum$	Summation operator.	$\sum_{i=1}^{4} A_i = A_1 + A_2 + A_3 + A_4$.
max	Maximum operator.	$\max_{t \in T} f(t)$ evaluates $f(t)$ over all $t \in T$ and returns the maximum value. $\max \boldsymbol{A}$ returns the largest value of the set $\boldsymbol{A}$, e.g., $\max\{1, 2, 3\} = 3$.
min	Minimum operator.	$\min_{t \in T} f(t)$ evaluates $f(t)$ over all $t \in T$ and returns the minimum value. $\min \boldsymbol{A}$ returns the smallest value of the set $\boldsymbol{A}$, e.g., $\min\{1, 2, 3\} = 1$.
$\times$	Multiplication operator.	
$\mathbb{I}\{A\}$	Indicator function returning 1 if condition A is true and 0 otherwise.	Defined in (5).
Time and Indexing		
t	Time measured in minutes relative to encounter start.	Negative values denote pre-encounter measurements, $t = 0$ marks encounter start, and positive values occur during the encounter.
T	Time window measured in minutes.	$T = [0, 1440)$ represents the first 24 hours following encounter start.
m_{resp}	Look-back window for respiratory related values	$[0, \infty)$
m_{vaso}	Look-back window for vasoactive medications	$[0, \infty)$
m_{bp}	Look-back window for blood pressure measurements	$[0, \infty)$
m_{lac}	Look-back window for lactate	$[0, \infty)$
m_{gcs}	Look-back window for GCS	$[0, \infty)$
m_{pupil}	Look-back window for Pupils	$[0, \infty)$
m_{coag}	Look-back window for coagulation related values	$[0, \infty)$
m_{endo}	Look-back window for endocrine related values	$[0, \infty)$
m_{immu}	Look-back window for immunologic related values	$[0, \infty)$
m_{hepatic}	Look-back window for hepatic related values	$[0, \infty)$
m_{renal}	Look-back window for renal related values	$[0, \infty)$

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Symbol	Meaning	Example or Notes
$\boldsymbol{m}$	Vector of look-back windows used for carry-forward operations.	
Observation History and Carry-Forward		
$\tau_X(t)$	Most recent observation time for variable X at or before time t.	$\tau_X(t)$ identifies when X was last measured prior to t. Defined in (2).
$\Delta_X(t)$	Staleness of measurement for variable X.	$\Delta_X(t)$ is the elapsed time since $\tau_X(t)$, or ∞ if X has not been observed. Defined in (3).
$\mathcal{C}[X](t;m)$	Carry-forward operator returning the most recent value of X within m minutes of time t.	Returns the last observed value of X if $\tau_X(t) \neq \text{NA}$ and $\Delta_X(t) \leq m$, otherwise NA. Defined in (4).
$\mathcal{D}\left[x\right](t;\cdot)$	Scoring operator: the point contribution of scoring entity x at time t.	When the common argument list $(\boldsymbol{m}, \text{Age})$ is used, arguments not relevant to x are ignored.
NA	Missing, unknown, or undefined value.	Returned when a required measurement is unavailable.
Interval-Level Status and Scores		
$\text{SI}(T)$	Suspected infection status during the interval T.	Equals 1 when the qualifying infectious test and anti-infective administration occur within T; otherwise 0.
$\text{ODSS}_d(T;\boldsymbol{m},\text{Age})$	Organ Dysfunction Summary Score for the organ-system set Ω_d.	Summarizes organ dysfunction over T independently of suspected infection status.
$\text{PSS}_d(T;\boldsymbol{m},\text{Age})$	Phoenix Sepsis Score for the organ-system set Ω_d.	Defined as $\text{SI}(T)\text{ODSS}_d(T;\boldsymbol{m},\text{Age})$.

Two points are especially important for reading the rest of the document. First, $:=$ is **not** the same as $=$. $:=$ means we are *defining* something. Second, NA means "missing/unknown/undefined" (e.g., a lab value was not measured), which is different from a numeric value like 0. By contrast, $\emptyset$ is the **empty set** (no elements), which we use in set-based definitions like $\max \emptyset = \text{NA}$.

For arithmetic, when NA is involved, the outcome is always NA, e.g., $\text{NA} + X := \text{NA}\ \forall X$. Table 2 explicitly defines how logical statements are interpreted when working with NA. When a value is missing, we avoid forcing it into "true" or "false." The rules below show how missingness propagates through logical statements. If either side of an "and/or" statement is unknown, the whole statement may be unknown unless the other side already determines the answer.

2.2 Function Definitions

This section defines how we turn time-stamped clinical data into a single value at any time t. In real EHR data, measurements are taken at irregular times. The definitions below formalize "what is the most recent value?" and "is it too old to trust?" so that scoring is reproducible.

For any time-indexed physiologic variable X, we write $X(t)$ for the value recorded at time t where t is the time, in minutes from the encounter start, i.e., $t = 0$ denotes encounter start, $t = 32$ is 32 minutes into the encounter, and $t = -12$ is 12 minutes before encounter start. The negative times represent measurements recorded before encounter start, such as EMS data.

Table 2: Logic with NA.

Statement	Value
'OR' statements	
NA $\vee$ NA	NA
NA $\vee$ true	true
NA $\vee$ false	NA
'And' statements	
NA $\wedge$ NA	NA
NA $\wedge$ true	NA
NA $\wedge$ false	false

The notation $X(t)$ assumes that X has a single value at each encounter time. Raw EHR extracts do not always satisfy this assumption. More than one row may exist for the same patient encounter, Phoenix input, and encounter-clock time, for example because values were received from multiple source tables or because an interface reported an amended value with the same timestamp.

Before the carry-forward definitions below are applied, duplicate raw rows for the same Phoenix input and same encounter-clock time are collapsed to one value:

$$X(t) \coloneqq g_X\left(\{x : x \text{ was reported for input } X \text{ at time } t\}\right), \tag{1}$$

where g_X is the input-specific duplicate-resolution function. The default duplicate-resolution functions are listed with each organ-system EHR input table below. Choosing a different g_X changes the operational definition of the prepared input stream.

For any variable X, define the last observation time prior to or at t as:

$$\tau_X(t) \coloneqq \max\left\{s \leq t : X(s) \in \boldsymbol{D}\right\}, \quad \max \emptyset = \text{NA}, \tag{2}$$

where $\boldsymbol{D}$ is a set, or interval, of valid values.

Define the staleness of the measurement:

$$\Delta_X(t) \coloneqq \begin{cases} t - \tau_X(t) & \tau_X(t) \neq \text{NA} \\ \infty & \text{otherwise.} \end{cases} \tag{3}$$

We define a staleness-limited carry-forward operator. This operator represents carry-forward when working retrospectively and a look-back limit when working prospectively or in real time. For $m \geq 0$ minutes,

$$\mathcal{C}\left[X\right](t; m) \coloneqq \begin{cases} X\left(\tau_X(t)\right) & \tau_X(t) \neq \text{NA} \wedge \Delta_X(t) \leq m \\ \text{NA} & \text{otherwise.} \end{cases} \tag{4}$$

We will use an indicator function that returns 1 or 0:

$$\mathbb{I}\left\{A\right\} \coloneqq \begin{cases} 1 & A \text{ is true} \\ 0 & A \text{ is false or NA.} \end{cases} \tag{5}$$

In words: $\tau_X(t)$ tells you when the last measurement happened, $\Delta_X(t)$ tells you how old it is, with unobserved values treated as infinitely stale, and $\mathcal{C}[X](t; m)$ returns that last value if it exists and is recent enough (within

m minutes); otherwise it returns "missing." This is how we prevent using values that are too old to reflect current physiology. The indicator $\mathbb{I}\{A\}$ then lets us convert logical statements (e.g., "lactate is at least 5") into 0/1 points for scoring. Because $\mathbb{I}\{\text{NA}\} = 0$, missing or unevaluable component criteria contribute zero points unless another available measurement establishes dysfunction. This is the minimum-score treatment of missing data used in the historical implementation.

For potentially incomplete time-indexed function f, extrema are evaluated over the available values of f. Define:

$$\mathcal{A}_f\left(T\right) \coloneqq \left\{f\left(t\right) : t \in T, f\left(t\right) \text{ is available}\right\}. \tag{6}$$

Then

$$\max_{t \in T} f\left(t\right) \coloneqq \begin{cases} \max \mathcal{A}_f\left(T\right) & \mathcal{A}_f\left(T\right) \neq \emptyset \\ \text{NA} & \mathcal{A}_f\left(T\right) = \emptyset, \end{cases} \tag{7}$$

and

$$\min_{t \in T} f\left(t\right) \coloneqq \begin{cases} \min \mathcal{A}_f\left(T\right) & \mathcal{A}_f\left(T\right) \neq \emptyset \\ \text{NA} & \mathcal{A}_f\left(T\right) = \emptyset. \end{cases} \tag{8}$$

3 Defining Organ System Dysfunction Scores

We assess eight organ systems: respiratory, cardiovascular, neurologic, coagulation, endocrine, immunologic, hepatic, and renal. Each subsection below lists the EHR inputs needed to compute that organ score, then shows how those inputs are transformed into constructed variables and point assignments. The goal is to make every step of the scoring reproducible: given the same time-stamped data, you should arrive at the same score.

The listed patient, laboratory, observation, event, and medication fields are as we expect them to appear in the electronic health record. These are inputs, not judgments about clinical severity. We define valid values for each variable. These valid-value ranges are intentionally broad (wider than clinically plausible values) to keep data validation separate from scoring.

3.1 Respiratory

The respiratory score summarizes oxygenation severity in the context of respiratory support. This subsection identifies the EHR inputs needed for oxygenation ratios and support indicators, defines the constructed PF ratio, SF ratio, invasive mechanical ventilation, and other respiratory support variables, and then maps those constructed variables to respiratory dysfunction points.

3.1.1 Data from the EHR

Several values from the Electronic Health Record capture oxygenation: FiO_2 (fraction of inspired oxygen), SpO_2 (pulse oximetry oxygen saturation), and PaO_2 (arterial oxygen pressure). The EHR also records the presence of mechanical or noninvasive respiratory support. Together, they allow us to quantify how well the patient is oxygenating and whether that oxygenation depends on advanced support. See Table 3.

Table 3: EHR Inputs for Respiratory Organ Dysfunction Scoring.

Variable	Description	Type	Units	Valid values	g_X
FiO_2	Fraction of Inspired Oxygen	Real	unitless	$[0.21, 1.00]$	max
SpO_2	Pulse oximetry oxygen saturation	Integer	unitless	$\{0, 1, \ldots, 100\}$	min
PaO_2	Arterial oxygen pressure	Real	mmHg	$[0, \infty)$	min
P_{aw}^{VENT}	Mean Airway Pressure, from a ventilator	Real	cm H_2O	$[0, \infty)$	max
P_{aw}^{HFOV}	Mean Airway Pressure, from a HFOV	Real	cm H_2O	$[0, \infty)$	max
P_{aw}^{PEEP}	Positive end-expiratory pressure	Real	cm H_2O	$[0, \infty)$	max
VENT	Indicator for invasive mechanical ventilation	Integer		$\{0, 1\}$	max
O_2Support	Any non-invasive oxygen support	Integer		$\{0, 1\}$	max

3.1.2 Constructed Variables

From the EHR values we construct several variables which are then used for the respiratory score. The constructed variables combine measurements that are not always recorded at the same time (e.g., SpO_2 and FiO_2) and apply clinical logic about timing and support. This keeps us from pairing an outdated FiO_2 with a more recent oxygen measurement or vice versa.

PF Ratio For determining the PF ratio at time t we consider look-back windows for both FiO_2 and PaO_2 and the relative look-back between the two. Specifically, the PFR is only valid when

$$\tau_{\mathrm{FiO_2}}(t) \leq \tau_{\mathrm{PaO_2}}(t) \quad \equiv \quad \Delta_{\mathrm{FiO_2}}(t) \geq \Delta_{\mathrm{PaO_2}}(t). \tag{9}$$

When both measurements have been observed, this is equivalent to $\Delta_{\mathrm{FiO_2}}(t) \geq \Delta_{\mathrm{PaO_2}}(t)$. Conceptually, this restriction is in place to protect from using a PaO_2 value which was taken *before* the most current FiO_2 value. That is, PFR is only valid if the PaO_2 value is reported at the same time, or after the FiO_2 is reported.

The PFR at time t is:

$$\mathrm{PFR}\left(t; m_{\mathrm{resp}}\right) := \begin{cases} \frac{\mathcal{C}[\mathrm{PaO_2}](t; m_{\mathrm{resp}})}{\mathcal{C}[\mathrm{FiO_2}](t; m_{\mathrm{resp}})} & \tau_{\mathrm{FiO_2}}(t) \leq \tau_{\mathrm{PaO_2}}(t) \\ \mathrm{NA} & \text{otherwise.} \end{cases} \tag{10}$$

With this definition, at time t, the PF ratio will either have a valid numeric value or be NA.

SF Ratio Like the PF ratio, we only consider the SF ratio valid if the SpO_2 value is reported at the same time as the FiO_2 or after.

$$\tau_{\mathrm{FiO_2}}(t) \leq \tau_{\mathrm{SpO_2}}(t) \quad \equiv \quad \Delta_{\mathrm{FiO_2}}(t) \geq \Delta_{\mathrm{SpO_2}}(t). \tag{11}$$

When both measurements have been observed, this is equivalent to $\Delta_{\mathrm{FiO_2}}(t) \geq \Delta_{\mathrm{SpO_2}}(t)$.

Additionally, the SF ratio is valid if the $SpO_2 \leq 97$. This requirement is based on the PODIUM [Bembea et al., 2022, Yehya et al., 2022] organ dysfunction scoring, and is an additional requirement on the pSOFA [Matics and Sanchez-Pinto, 2017] cut points in the scoring.

$$\mathrm{SFR}\left(t; m_{\mathrm{resp}}\right) \coloneqq \begin{cases} \frac{\mathcal{C}[\mathrm{SpO}_2](t;m_{\mathrm{resp}})}{\mathcal{C}[\mathrm{FiO}_2](t;m_{\mathrm{resp}})} & \tau_{\mathrm{FiO}_2}(t) \leq \tau_{\mathrm{SpO}_2}(t) \wedge \mathcal{C}\left[\mathrm{SpO}_2\right](t; m_{\mathrm{resp}}) \leq 97 \\ \mathrm{NA} & \text{otherwise.} \end{cases} \tag{12}$$

Invasive Mechanical Ventilation We operationalize the definition for invasive mechanical ventilation based in part of differences in the EHR records we had access to. In some sets, a simple VENT indicator variable was provided. In other EHR sets we had mean airway pressure from either a ventilator or high-frequency oscillatory ventilation (HFOV). Additionally, we had positive end-expiratory pressure (PEEP) to consider as well.

We defined invasive mechanical ventilation (IMV) as an indicator based on four possible conditions.

$$\mathrm{IMV}\left(t; m_{\mathrm{resp}}\right) \coloneqq \mathbb{I}\left\{\bigvee_{i=1}^{4} A_i\left(t; m_{\mathrm{resp}}\right)\right\} \tag{13}$$

where

$$\begin{aligned} A_1\left(t; m_{\mathrm{resp}}\right) &\coloneqq \mathcal{C}\left[\mathrm{VENT}\right]\left(t; m_{\mathrm{resp}}\right) = 1, \\ A_2\left(t; m_{\mathrm{resp}}\right) &\coloneqq \mathcal{C}\left[\mathrm{P}_{\mathrm{aw}}^{\mathrm{VENT}}\right]\left(t; m_{\mathrm{resp}}\right) > 0, \\ A_3\left(t; m_{\mathrm{resp}}\right) &\coloneqq \mathcal{C}\left[\mathrm{P}_{\mathrm{aw}}^{\mathrm{HFOV}}\right]\left(t; m_{\mathrm{resp}}\right) > 0, \text{ and} \\ A_4\left(t; m_{\mathrm{resp}}\right) &\coloneqq \mathcal{C}\left[\mathrm{P}_{\mathrm{aw}}^{\mathrm{PEEP}}\right]\left(t; m_{\mathrm{resp}}\right) > 3. \end{aligned} \tag{14}$$

The PEEP limit of 3 cm H_2O is used to limit false positives while limiting false negatives. Similar limits have been used in other criteria. For example, the lung injury score [Martin and Bernard, 2001] uses a PEEP of 5 cm H_2O for 0 points and 6-8 cm H_2O. Further, Martin and Bernard [2001] states a PEEP of 5 cm H_2O and $\mathrm{FiO}_2 < 0.5$ as part of the criteria for identifying patients ready to accept a trial of spontaneous breathing.

Other Respiratory Support Extend the definitions in (14) to include

$$\begin{aligned} A_5\left(t; m_{\mathrm{resp}}\right) &\coloneqq \mathcal{C}\left[\mathrm{FiO}_2\right]\left(t; m_{\mathrm{resp}}\right) > 0.21, \\ A_6\left(t; m_{\mathrm{resp}}\right) &\coloneqq \mathcal{C}\left[\mathrm{O}_2\mathrm{Support}\right]\left(t; m_{\mathrm{resp}}\right) = 1. \end{aligned} \tag{15}$$

Then other respiratory support is defined by (14) and (15) as

$$\mathrm{ORS}\left(t; m_{\mathrm{resp}}\right) \coloneqq \mathbb{I}\left\{\bigvee_{i=1}^{6} A_i\left(t; m_{\mathrm{resp}}\right)\right\} \tag{16}$$

3.1.3 Respiratory Organ Dysfunction Score

In the development of Phoenix, we used pSOFA respiratory scoring[Matics and Sanchez-Pinto, 2017]. The pSOFA respiratory scoring does not explicitly require respiratory support to score 1 point, however, for the Phoenix definition the requirement of some respiratory support was made explicit. It is extremely unlikely that a patient with SFR or PFR values sufficiently low enough to score at least one point for respiratory dysfunction would not have at the very least supplemental oxygen.

$$\mathcal{D}\left[\mathrm{Resp}\right]\left(t; m_{\mathrm{resp}}\right) \coloneqq \mathrm{ORS}\left(t; m_{\mathrm{resp}}\right) B_1\left(t; m_{\mathrm{resp}}\right) + \mathrm{IMV}\left(t; m_{\mathrm{resp}}\right) \sum_{i=2}^{3} B_i\left(t; m_{\mathrm{resp}}\right) \tag{17}$$

where

$$
\begin{aligned}
B_1\left(t;m_{\text{resp}}\right) &\coloneqq Q\left(t;m_{\text{resp}},\delta_{\text{PaO}_2,\text{SpO}_2}\right)\mathbb{I}\left\{\text{PFR}\left(t;m_{\text{resp}}\right)<400\right\}\\
&\quad+\left(1-Q\left(t;m_{\text{resp}},\delta_{\text{PaO}_2,\text{SpO}_2}\right)\right)\mathbb{I}\left\{\text{SFR}\left(t;m_{\text{resp}}\right)<292\right\},\\
B_2\left(t;m_{\text{resp}}\right) &\coloneqq Q\left(t;m_{\text{resp}},\delta_{\text{PaO}_2,\text{SpO}_2}\right)\mathbb{I}\left\{\text{PFR}\left(t;m_{\text{resp}}\right)<200\right\}\\
&\quad+\left(1-Q\left(t;m_{\text{resp}},\delta_{\text{PaO}_2,\text{SpO}_2}\right)\right)\mathbb{I}\left\{\text{SFR}\left(t;m_{\text{resp}}\right)<220\right\},\\
B_3\left(t;m_{\text{resp}}\right) &\coloneqq Q\left(t;m_{\text{resp}},\delta_{\text{PaO}_2,\text{SpO}_2}\right)\mathbb{I}\left\{\text{PFR}\left(t;m_{\text{resp}}\right)<100\right\}\\
&\quad+\left(1-Q\left(t;m_{\text{resp}},\delta_{\text{PaO}_2,\text{SpO}_2}\right)\right)\mathbb{I}\left\{\text{SFR}\left(t;m_{\text{resp}}\right)<148\right\},
\end{aligned}
\tag{18}
$$

and

$$
Q\left(t;m_{\text{resp}},\delta_{\text{PaO}_2,\text{SpO}_2}\right)\coloneqq
\begin{cases}
1 & \text{PFR}\left(t;m_{\text{resp}}\right)\neq\text{NA}\wedge\text{SFR}\left(t;m_{\text{resp}}\right)=\text{NA}\\
0 & \text{PFR}\left(t;m_{\text{resp}}\right)=\text{NA}\wedge\text{SFR}\left(t;m_{\text{resp}}\right)\neq\text{NA}\\
1 & \begin{array}{l}\text{PFR}\left(t;m_{\text{resp}}\right)\neq\text{NA}\wedge\text{SFR}\left(t;m_{\text{resp}}\right)\neq\text{NA}\wedge\\ \Delta_{\text{PaO}_2}\left(t\right)\leq\Delta_{\text{SpO}_2}\left(t\right)+\delta_{\text{PaO}_2,\text{SpO}_2}\end{array}\\
0 & \begin{array}{l}\text{PFR}\left(t;m_{\text{resp}}\right)\neq\text{NA}\wedge\text{SFR}\left(t;m_{\text{resp}}\right)\neq\text{NA}\wedge\\ \Delta_{\text{PaO}_2}\left(t\right)>\Delta_{\text{SpO}_2}\left(t\right)+\delta_{\text{PaO}_2,\text{SpO}_2}\end{array}\\
0 & \text{PFR}\left(t;m_{\text{resp}}\right)=\text{NA}\wedge\text{SFR}\left(t;m_{\text{resp}}\right)=\text{NA}
\end{cases}
\tag{19}
$$

The parameter $\delta_{\text{PaO}_2,\text{SpO}_2}$ is a relative staleness value, in minutes, between the PaO_2 and SpO_2 measurement times. When both the PFR and SFR are known at time t within the carry-forward (look-back) window m_{resp} then the PFR is taken preferentially over SFR unless SFR is sufficiently younger than the PFR.

The valid values for the respiratory organ dysfunction score are

$$
\mathcal{D}\left[\text{Resp}\right]\left(t;m_{\text{resp}}\right)\in\left\{0,1,2,3\right\}\quad\forall m_{\text{resp}},t. \tag{20}
$$

3.2 Cardiovascular

The cardiovascular score has three components, vasoactive medications, mean arterial pressure, and lactate. A score of 0, 1, or 2 points is possible for each of the components resulting in a total score of 0 to 6 points.

3.2.1 Data from the EHR

The input data from the EHR needed for the cardiovascular dysfunction scores are vasoactive medications (as indicators of pharmacologic cardiovascular support), blood pressure measurements (from cuff or arterial line), lactate, and age (for age-specific mean arterial pressure thresholds). See Table 4.

3.2.2 Constructed Variable

We build three components: vasoactive medication use, mean arterial pressure, and lactate. Each component contributes 0–2 points, and together they sum to the cardiovascular score. The definitions below formalize how we combine multiple data sources (e.g., cuff vs arterial line) and how we handle timing.

Table 4: EHR Inputs for Cardiovascular Organ Dysfunction Scoring.

Variable	**Description**	**Type**	**Units**	**Valid values**	g_X
Dobutamine	Indicator of systemic use of vasoactive medication	Integer	–	$\{0, 1\}$	max
Dopamine	Indicator of systemic use of vasoactive medication	Integer	–	$\{0, 1\}$	max
Epinephrine	Indicator of systemic use of vasoactive medication	Integer	–	$\{0, 1\}$	max
Milrinone	Indicator of systemic use of vasoactive medication	Integer	–	$\{0, 1\}$	max
Norepinephrine	Indicator of systemic use of vasoactive medication	Integer	–	$\{0, 1\}$	max
Vasopressin	Indicator of systemic use of vasoactive medication	Integer	–	$\{0, 1\}$	max
Lactate	Lactate	Real	mmol / L	$[0, \infty)$	max
SBP_c	systolic blood pressure reported from a cuff	Real	mmHg	$[0, \infty)$	max
DBP_c	diastolic blood pressure reported from a cuff	Real	mmHg	$[1, 200]$	max
MAP_c	mean arterial pressure reported from a cuff	Real	mmHg	$[0, \infty)$	max
SBP_a	systolic blood pressure reported from arterial line	Real	mmHg	$[0, \infty)$	max
DBP_a	diastolic blood pressure reported from arterial line	Real	mmHg	$[1, 200]$	max
MAP_a	mean arterial pressure reported from arterial line	Real	mmHg	$[0, \infty)$	max
Age	Patient's age	Real	months	$[0, 216)$	min

Vasoactive Medications Each of the six vasoactive medications,

$$\boldsymbol{V} \coloneqq \{\text{Dobutamine}, \text{Dopamine}, \text{Epinephrine}, \text{Milrinone}, \text{Norepinephrine}, \text{Vasopressin}\}, \tag{21}$$

are to be assessed as 0/1 indicators. Either the patient is receiving systemic vasoactive medication for the purpose of cardiovascular support or they are not. Medications such as epinephrine are used in many different situations for different purposes, it is important that the input data is an indicator of the use of the medication systemically for cardiovascular support.

We define the contribution to the cardiovascular score from vasoactive medications as the count of the number of vasoactive medications received, regardless of dose, with a cap of 2 points.

$$\mathcal{D}\left[\text{Vasos}\right](t; m_{\text{vaso}}) \coloneqq \min\left\{2, \sum_{v \in \boldsymbol{V}} \mathbb{I}\left\{\mathcal{C}[v](t; m_{\text{vaso}}) = 1\right\}\right\}. \tag{22}$$

The valid values for the contribution to the cardiovascular score from vasoactive medications are:

$$\mathcal{D}\left[\text{Vasos}\right](t; m_{\text{vaso}}) \in \{0, 1, 2\} \quad \forall m_{\text{vaso}}, t \tag{23}$$

This was based on the use of the vasoactive inotrope score (VIS)[Haque et al., 2015]. The VIS used the same six vasoactive medications and based the score on the dosage. During the development of the Phoenix criteria we found some lower resourced sites did not have all six medications. As such, patients at these lower resourced sites had a ceiling on the severity of cardiovascular dysfunction based on VIS. By ignoring the dosage and just relying on the fact that the patient was receiving the medication, the Phoenix criteria was determined to have greater utility across high-, medium-, and low- resourced sites.

Mean Arterial Pressure (MAP) Mean arterial pressure (MAP) is non-trivial to define. We have limits based on age (in months) and a hierarchy for the preferable source of information used to define the MAP. In practice, MAP may be recorded directly or inferred from systolic and diastolic pressures. We prioritize youngest measurements, within a tolerance, and prioritize arterial line measurements when available because they are more reliable in unstable patients, then fall back to cuff measurements.

Use the youngest, within a window, mean arterial pressure.

$$\begin{aligned}
\widehat{\text{MAP}}_1(t; m_{\text{bp}}) &\coloneqq \mathcal{C}[\text{MAP}_\text{a}](t; m_{\text{bp}}), \\
\widehat{\text{MAP}}_2(t; m_{\text{bp}}) &\coloneqq \begin{cases} \frac{1}{3}\mathcal{C}[\text{SBP}_\text{a}](t; m_{\text{bp}}) + \frac{2}{3}\mathcal{C}[\text{DBP}_\text{a}](t; m_{\text{bp}}) & \begin{array}{l}\mathcal{C}[\text{SBP}_\text{a}](t; m_{\text{bp}}) \neq \text{NA} \wedge \\ \mathcal{C}[\text{DBP}_\text{a}](t; m_{\text{bp}}) \neq \text{NA} \wedge \\ |\Delta_{\text{SBP}_\text{a}}(t) - \Delta_{\text{DBP}_\text{a}}(t)| \leq \delta_{\text{sdbp}}\end{array} \\ \text{NA} & \text{otherwise,} \end{cases} \\
\widehat{\text{MAP}}_3(t; m_{\text{bp}}) &\coloneqq \mathcal{C}[\text{MAP}_\text{c}](t; m_{\text{bp}}), \text{ and} \\
\widehat{\text{MAP}}_4(t; m_{\text{bp}}) &\coloneqq \begin{cases} \frac{1}{3}\mathcal{C}[\text{SBP}_\text{c}](t; m_{\text{bp}}) + \frac{2}{3}\mathcal{C}[\text{DBP}_\text{c}](t; m_{\text{bp}}) & \begin{array}{l}\mathcal{C}[\text{SBP}_\text{c}](t; m_{\text{bp}}) \neq \text{NA} \wedge \\ \mathcal{C}[\text{DBP}_\text{c}](t; m_{\text{bp}}) \neq \text{NA} \wedge \\ |\Delta_{\text{SBP}_\text{c}}(t) - \Delta_{\text{DBP}_\text{c}}(t)| \leq \delta_{\text{sdbp}}\end{array} \\ \text{NA} & \text{otherwise,} \end{cases}
\end{aligned} \tag{24}$$

where δ_{sdbp} is the maximum allowed time difference, in minutes, between systolic and diastolic pressures used to estimate MAP. The effective staleness of each MAP candidate is

$$
\begin{aligned}
\eta_1(t) &\coloneqq \begin{cases} \Delta_{\mathrm{MAP_a}}(t) & \widehat{\mathrm{MAP}}_1(t; m_{\mathrm{bp}}) \neq \mathrm{NA} \\ \infty & \text{otherwise,} \end{cases} \\
\eta_2(t) &\coloneqq \begin{cases} \min\{\Delta_{\mathrm{SBP_a}}(t), \Delta_{\mathrm{DBP_a}}(t)\} & \widehat{\mathrm{MAP}}_2(t; m_{\mathrm{bp}}) \neq \mathrm{NA} \\ \infty & \text{otherwise,} \end{cases} \\
\eta_3(t) &\coloneqq \begin{cases} \Delta_{\mathrm{MAP_c}}(t) & \widehat{\mathrm{MAP}}_3(t; m_{\mathrm{bp}}) \neq \mathrm{NA} \\ \infty & \text{otherwise,} \end{cases} \\
\eta_4(t) &\coloneqq \begin{cases} \min\{\Delta_{\mathrm{SBP_c}}(t), \Delta_{\mathrm{DBP_c}}(t)\} & \widehat{\mathrm{MAP}}_4(t; m_{\mathrm{bp}}) \neq \mathrm{NA} \\ \infty & \text{otherwise.} \end{cases}
\end{aligned} \tag{25}
$$

Then, let δ_{MAP} denote the MAP candidate freshness value, in minutes, used to decide when candidate sources are treated as having similar freshness and the source hierarchy should break the tie.

$$
\mathrm{MAP}^{\star}(t; m_{\mathrm{bp}}) \coloneqq \begin{cases}
\widehat{\mathrm{MAP}}_1(t; m_{\mathrm{bp}}) & \begin{array}{l} \eta_1(t) < \infty \quad \wedge \\ \eta_1(t) \leq \min\{\eta_2(t), \eta_3(t), \eta_4(t)\} + \delta_{\mathrm{MAP}} \end{array} \\[2ex]
\widehat{\mathrm{MAP}}_2(t; m_{\mathrm{bp}}) & \begin{array}{l} \eta_2(t) < \infty \quad \wedge \\ \eta_2(t) < \eta_1(t) + \delta_{\mathrm{MAP}} \quad \wedge \\ \eta_2(t) \leq \min\{\eta_3(t), \eta_4(t)\} + \delta_{\mathrm{MAP}} \end{array} \\[2ex]
\widehat{\mathrm{MAP}}_3(t; m_{\mathrm{bp}}) & \begin{array}{l} \eta_3(t) < \infty \quad \wedge \\ \eta_3(t) < \min\{\eta_1(t), \eta_2(t)\} + \delta_{\mathrm{MAP}} \quad \wedge \\ \eta_3(t) \leq \eta_4(t) + \delta_{\mathrm{MAP}} \end{array} \\[2ex]
\widehat{\mathrm{MAP}}_4(t; m_{\mathrm{bp}}) & \begin{array}{l} \eta_4(t) < \infty \quad \wedge \\ \eta_4(t) < \min\{\eta_1(t), \eta_2(t), \eta_3(t)\} + \delta_{\mathrm{MAP}} \end{array} \\[2ex]
\mathrm{NA} & \text{otherwise.}
\end{cases} \tag{26}
$$

Plain English Goal: At one score time, choose one MAP value to use for Phoenix scoring. MAP can come from four places:

1. Arterial-line MAP measured directly.
2. Arterial-line MAP calculated from arterial SBP and DBP.
3. Cuff MAP measured directly.
4. Cuff MAP calculated from cuff SBP and DBP.

The limits for the MAP are based on age (in months). Age-based thresholds reflect pediatric physiology:

younger patients have lower expected MAPs, so the cut-points increase with age. Let

$$\theta_1(\text{Age}) \coloneqq \begin{cases} 31 & \text{Age} \in [0, 1) \\ 39 & \text{Age} \in [1, 12) \\ 44 & \text{Age} \in [12, 24) \\ 45 & \text{Age} \in [24, 60) \\ 49 & \text{Age} \in [60, 144) \\ 52 & \text{Age} \in [144, 216), \end{cases} \tag{27}$$

and

$$\theta_2(\text{Age}) \coloneqq \begin{cases} 17 & \text{Age} \in [0, 1) \\ 25 & \text{Age} \in [1, 12) \\ 31 & \text{Age} \in [12, 24) \\ 32 & \text{Age} \in [24, 60) \\ 36 & \text{Age} \in [60, 144) \\ 38 & \text{Age} \in [144, 216). \end{cases} \tag{28}$$

Then, the part of the cardiovascular score based on MAP is:

$$\mathcal{D}\left[\text{MAP}\right](t; m_{\text{bp}}, \text{Age}) \coloneqq \mathbb{I}\left\{\text{MAP}^{\star}(t; m_{\text{bp}}) < \theta_2(\text{Age})\right\} + \mathbb{I}\left\{\text{MAP}^{\star}(t; m_{\text{bp}}) < \theta_1(\text{Age})\right\}. \tag{29}$$

The valid values for this component of the cardiovascular dysfunction score are

$$\mathcal{D}\left[\text{MAP}\right](t; m_{\text{bp}}, \text{Age}) \in \{0, 1, 2\} \quad \forall m_{\text{bp}}, t. \tag{30}$$

Lactate Lactate is treated as a marker of impaired perfusion and severity of shock. Higher lactate thresholds contribute more points. The definition for lactate used in the cardiovascular dysfunction score is:

$$\mathcal{D}\left[\text{Lactate}\right](t; m_{\text{lac}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{Lactate}](t; m_{\text{lac}}) \geq 5\right\} + \mathbb{I}\left\{\mathcal{C}[\text{Lactate}](t; m_{\text{lac}}) \geq 11\right\}. \tag{31}$$

That is, one point for a lactate value of at least 5 mmol/L; two points for a lactate value of at least 11 mmol/L. The valid values for the contribution of lactate to the cardiovascular dysfunction score are:

$$\mathcal{D}\left[\text{Lactate}\right](t; m_{\text{lac}}) \in \{0, 1, 2\} \quad \forall m_{\text{lac}}, t. \tag{32}$$

3.2.3 Cardiovascular Dysfunction Score

The cardiovascular dysfunction is based on (22), (29), and (31).

$$\mathcal{D}\left[\text{Card}\right](t; m_{\text{vaso}}, m_{\text{bp}}, m_{\text{lac}}, \text{Age}) \coloneqq \sum_{x \in \boldsymbol{C}_{\text{card}}} \mathcal{D}\left[x\right](t; \boldsymbol{m}, \text{Age}) \tag{33}$$

where

$$\boldsymbol{C}_{\text{card}} \coloneqq \{\text{Vasos}, \text{MAP}, \text{Lactate}\}. \tag{34}$$

The valid values for the cardiovascular dysfunction score are

$$\mathcal{D}\left[\text{Card}\right](t; m_{\text{vaso}}, m_{\text{bp}}, m_{\text{lac}}, \text{Age}) \in \{0, 1, 2, 3, 4, 5, 6\} \quad \forall \boldsymbol{m}, t. \tag{35}$$

Table 5: EHR Inputs for Neurological Organ Dysfunction Scoring.

Variable	Description	Type	Units	Valid values	g_X
GCS_motor	Glasgow Coma Motor Score	Integer	unitless	$\{1,2,3,4,5,6\}$	max
GCS_eye	Glasgow Coma Eye Score	Integer	unitless	$\{1,2,3,4\}$	max
GCS_verbal	Glasgow Coma Verbal Score	Integer	unitless	$\{1,2,3,4,5\}$	max
GCS_total	Total Glasgow Coma Score	Integer	unitless	$\{3,4,\ldots,15\}$	max
Pupils	Bilaterally fixed pupils	Integer		$\{0,1\}$	max

3.3 Neurological

The neurological score is based on the Glasgow Coma score (GCS) and whether or not the patient has bilaterally fixed pupils. These capture level of consciousness and brainstem function, respectively.

3.3.1 Data from the EHR

The input data from the EHR needed for neurological dysfunction scoring are the Glasgow Coma Score components, the total Glasgow Coma Score when available, and the presence of bilaterally fixed pupils. See Table 5.

3.3.2 Constructed Variable

We derive two variables: a time-aligned GCS total (which may need to be reconstructed from components) and a binary indicator for fixed pupils. This helps address common EHR issues where component scores are updated without the total.

GCS In general the GCS total is the sum of the three components, eye, verbal, and motor. In the development of the Phoenix criteria we found that there were cases where one or two of the components would be updated but the other component(s) and the total were not updated. When this occurred we chose to assume it was because there was no change in the other component scores. This rule prevents missing totals from artificially lowering the score when only one component is updated.

$$\text{GCS}\left(t; m_\text{gcs}\right) \coloneqq \begin{cases} \mathcal{C}[\text{GCS}_\text{total}](t; m_\text{gcs}) & \Delta_{\text{GCS}_\text{total}}(t) < \min_{i \in \boldsymbol{G}} \Delta_{\text{GCS}_\text{i}}(t) \\ \sum_{i \in \boldsymbol{G}} \mathcal{C}[\text{GCS}_\text{i}](t; \infty) & \begin{array}{l} \left[\Delta_{\text{GCS}_\text{total}}(t) = \infty \vee \min\limits_{i \in \boldsymbol{G}} \Delta_{\text{GCS}_\text{i}}(t) \leq \Delta_{\text{GCS}_\text{total}}(t)\right] \wedge \\ \max\limits_{i \in \boldsymbol{G}} \Delta_{\text{GCS}_\text{i}}(t) < \infty \wedge \\ \min\limits_{i \in \boldsymbol{G}} \Delta_{\text{GCS}_\text{i}}(t) \leq m_\text{gcs} \end{array} \\ \text{NA} & \text{otherwise,} \end{cases} \tag{36}$$

where

$$\boldsymbol{G} \coloneqq \{\text{motor}, \text{eye}, \text{verbal}\}. \tag{37}$$

The use of $\mathcal{C}[\cdot](t; \infty)$ means "carry forward the most recent value with no time limit." In practice, this is only applied when each of the three component scores has been observed at least once. If no total GCS has been observed, this component-sum rule applies once all three components have been observed within the GCS look-back window. If the component sum and reported total GCS have the same effective timestamp, the component sum is used. So if the latest eye, verbal, and motor scores were recorded at different times, we take each component's most recent value and sum them to form a consistent total, even if those component timestamps are not identical.

The score contribution from GCS is

$$\mathcal{D}\left[\text{GCS}\right]\left(t; m_{\text{gcs}}\right) \coloneqq \mathbb{I}\left\{GCS\left(t; m_{\text{gcs}}\right) \leq 10\right\} \tag{38}$$

Pupils This is treated as a severe neurologic finding and can dominate the neurologic score. The bilaterally fixed pupils are defined as:

$$\mathcal{D}\left[\text{Pupils}\right]\left(t; m_{\text{pupil}}\right) \coloneqq 2 \times \mathbb{I}\left\{\mathcal{C}[Pupils](t; m_{\text{pupil}}) = 1\right\}. \tag{39}$$

3.3.3 Neurological Organ Dysfunction Score

The neurological component on the score is:

$$\mathcal{D}\left[\text{Neuro}\right]\left(t; m_{\text{gcs}}, m_{\text{pupil}}\right) \coloneqq \max\left\{\mathcal{D}\left[\text{GCS}\right]\left(t; m_{\text{gcs}}\right), \mathcal{D}\left[\text{Pupils}\right]\left(t; m_{\text{pupil}}\right)\right\}. \tag{40}$$

3.4 Coagulation

The coagulation score uses common laboratory markers of clotting and fibrinolysis. Each marker contributes at most one point, and the total is capped at 2. This keeps the score sensitive to abnormalities without overweighting multiple abnormal labs that can occur together in the same patient.

For readers without a clinical background: Clotting (coagulation) is the body's process for stopping bleeding by forming fibrin clots. Fibrinolysis is the counter-process that breaks down those clots once they are no longer needed. INR is a standardized measure of how long it takes blood to clot; higher INR means slower clotting and can reflect coagulopathy, liver dysfunction, or anticoagulation. D-dimer is a breakdown product of cross-linked fibrin, so elevated levels suggest recent or ongoing clot formation and breakdown, though it is not specific to any single disease.

In severe infection, clotting can be activated even without trauma. Inflammatory signals and endothelial injury can trigger coagulation pathways, leading to micro-clot formation in small vessels. At the same time, fibrinolysis can be overwhelmed or dysregulated. In extreme cases this becomes disseminated intravascular coagulation (DIC), where clotting and bleeding risk coexist.

3.4.1 Data from the EHR

The input data from the EHR needed for coagulation dysfunction scoring are platelets, INR, D-dimer, and fibrinogen. See Table 6.

3.4.2 Constructed Variable

There are no variables to construct.

Table 6: EHR Inputs for Coagulation Organ Dysfunction Scoring.

Variable	Description	Type	Units	Valid values	g_X
Platelets	Platelets	Real	1000 / μL	$[0, \infty)$	max
INR	INR	Real	unitless	$[0, \infty)$	max
DDimer	D-Dimer	Real	mg/L (FEU)	$[0, \infty)$	max
Fibrinogen	Fibrinogen	Real	mg/dL	$[0, \infty)$	max

Table 7: EHR Inputs for Endocrine Organ Dysfunction Scoring.

Variable	Description	Type	Units	Valid values	g_X
Glucose	Blood Glucose	Real	mg/dL	$[0, \infty)$	max

3.4.3 Coagulation Organ Dysfunction Score

Each indicator below tests whether a lab value crosses a clinically relevant threshold. The score sums how many thresholds are met, then caps at 2 points.

Define:

$$\mathcal{D}\left[\text{Platelets}\right](t; m_{\text{coag}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{Platelets}]\,(t; m_{\text{coag}}) < 100\right\}, \tag{41}$$

$$\mathcal{D}\left[\text{INR}\right](t; m_{\text{coag}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{INR}]\,(t; m_{\text{coag}}) > 1.3\right\}, \tag{42}$$

$$\mathcal{D}\left[\text{DDimer}\right](t; m_{\text{coag}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{DDimer}]\,(t; m_{\text{coag}}) > 2\right\}, \tag{43}$$

and

$$\mathcal{D}\left[\text{Fibrinogen}\right](t; m_{\text{coag}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{Fibrinogen}]\,(t; m_{\text{coag}}) < 100\right\}. \tag{44}$$

Then, the coagulation score is:

$$\mathcal{D}\left[\text{Coag}\right](t; m_{\text{coag}}) \coloneqq \min\left\{2, \sum_{x \in \boldsymbol{C}_{\text{coag}}} \mathcal{D}\left[x\right](t; m_{\text{coag}})\right\}, \tag{45}$$

where

$$\boldsymbol{C}_{\text{coag}} \coloneqq \{\text{Platelets}, \text{INR}, \text{DDimer}, \text{Fibrinogen}\}. \tag{46}$$

3.5 Endocrine

The endocrine score is part of the Phoenix-8 scoring system.

3.5.1 Data from the EHR

There is only one element from the EHR used in the endocrine dysfunction assessment: glucose. See Table 7.

3.5.2 Constructed Variables

There are no constructed variables to define.

Table 8: EHR Inputs for Immunologic Organ Dysfunction Scoring.

Variable	Description	Type	Units	Valid values	g_X
ANC	Absolute neutrophil count	Real	1000 cells / mm^3	$[0, \infty)$	min
ALC	Absolute lymphocyte count	Real	1000 cells / mm^3	$[0, \infty)$	min

3.5.3 Endocrine Organ Dysfunction Score

The endocrine score converts glucose into a binary dysfunction indicator. Values outside the normal operating range contribute one point, while values within the range contribute zero points.

$$\mathcal{D}\left[\text{Endo}\right](t; m_{\text{endo}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{Glucose}]\left(t; m_{\text{endo}}\right) < 50 \vee \mathcal{C}[\text{Glucose}]\left(t; m_{\text{endo}}\right) > 150\right\} \tag{47}$$

The valid values for the endocrine score are

$$\mathcal{D}\left[\text{Endo}\right](t; m_{\text{endo}}) \in \{0, 1\}, \quad \forall m_{\text{endo}}, t. \tag{48}$$

3.6 Immunologic

The immunologic score is part of the Phoenix-8 scoring system.

3.6.1 Data from the EHR

Two values are required from the EHR: Absolute neutrophil count (ANC) and absolute lymphocyte count (ALC). Both are expected in units of 1000 cells per cubic millimeter. See Table 8.

3.6.2 Constructed Variables

There are no constructed variables for the immunologic score.

3.6.3 Immunologic Organ Dysfunction Score

The immunologic score converts low absolute lymphocyte count and low absolute neutrophil count into component indicators. The organ score is positive if either immune-cell count crosses its threshold. Define

$$\mathcal{D}\left[\text{ALC}\right](t; m_{\text{immu}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{ALC}]\left(t; m_{\text{immu}}\right) < 1.00\right\}, \tag{49}$$

and

$$\mathcal{D}\left[\text{ANC}\right](t; m_{\text{immu}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{ANC}]\left(t; m_{\text{immu}}\right) < 0.50\right\}. \tag{50}$$

Then

$$\mathcal{D}\left[\text{Immu}\right](t; m_{\text{immu}}) \coloneqq \max\left\{\mathcal{D}\left[\text{ALC}\right](t; m_{\text{immu}}), \mathcal{D}\left[\text{ANC}\right](t; m_{\text{immu}})\right\}. \tag{51}$$

The valid values for the immunologic score are

$$\mathcal{D}\left[\text{Immu}\right](t; m_{\text{immu}}) \in \{0, 1\}, \quad \forall m_{\text{immu}}, t. \tag{52}$$

Table 9: EHR Inputs for Hepatic Organ Dysfunction Scoring.

Variable	Description	Type	Units	Valid values	g_X
Bilirubin	Total bilirubin	Real	mg/dL	$[0, \infty)$	max
ALT	Alanine aminotransferase	Real	IU / L	$[0, \infty)$	max

3.7 Hepatic

The hepatic score is part of the Phoenix-8 scoring system. It uses total bilirubin and alanine aminotransferase (ALT) as markers of hepatic injury or dysfunction and maps them to a single binary organ dysfunction score.

3.7.1 Data from the EHR

Two values are required from the EHR: total bilirubin reported in units of mg/dL and ALT reported in units of IU/L. See Table 9.

3.7.2 Constructed Variables

There are no constructed variables for the hepatic organ dysfunction assessment.

3.7.3 Hepatic Organ Dysfunction Score

As with endocrine and immunologic scores, the hepatic score can be a 0 or 1 from either the bilirubin or ALT values.

Define:

$$\mathcal{D}\left[\text{Bilirubin}\right](t; m_{\text{hepatic}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{Bilirubin}]\,(t; m_{\text{hepatic}}) \geq 4\right\}, \tag{53}$$

and

$$\mathcal{D}\left[\text{ALT}\right](t; m_{\text{hepatic}}) \coloneqq \mathbb{I}\left\{\mathcal{C}[\text{ALT}]\,(t; m_{\text{hepatic}}) > 102\right\}. \tag{54}$$

Then

$$\mathcal{D}\left[\text{Hepatic}\right](t; m_{\text{hepatic}}) \coloneqq \max\left\{\mathcal{D}\left[\text{Bilirubin}\right](t; m_{\text{hepatic}}), \mathcal{D}\left[\text{ALT}\right](t; m_{\text{hepatic}})\right\}. \tag{55}$$

The valid values for the hepatic score are

$$\mathcal{D}\left[\text{Hepatic}\right](t; m_{\text{hepatic}}) \in \{0, 1\}, \quad \forall m_{\text{hepatic}}, t. \tag{56}$$

3.8 Renal

The renal score is part of the Phoenix-8 scoring system.

3.8.1 Data from the EHR

The patient's age (at admission, in months) and their creatinine levels (in mg/dL) are required to assess renal dysfunction. Age is also required for assessment of cardiovascular dysfunction as part of the Phoenix

Table 10: EHR Inputs for Renal Organ Dysfunction Scoring.

Variable	**Description**	**Type**	**Units**	**Valid values**	g_X
Creatinine	Creatinine	Real	mg / dL	$[0, \infty)$	max
Age	Patient's age	Real	months	$[0, 216)$	min

scoring and thus the only additional variable to collect to extend to Phoenix-8 from Phoenix for the renal dysfunction score is the creatinine. See Table 10.

3.8.2 Constructed Variables

There are no constructed variables required for the assessment of renal dysfunction.

3.8.3 Renal Organ Dysfunction Score

The renal dysfunction score is based on an age adjusted creatinine level (57). The cut points for the age ranges are the same as used in the cardiovascular scoring, (27) and (28).

$$\mathcal{D}\left[\text{Renal}\right](t; m_{\text{renal}}, \text{Age}) \coloneqq \begin{cases} 1 & \text{Age} \in [0, 1) \wedge \mathcal{C}[\text{Creatinine}]\left(t; m_{\text{renal}}\right) \geq 0.8 \\ 1 & \text{Age} \in [1, 12) \wedge \mathcal{C}[\text{Creatinine}]\left(t; m_{\text{renal}}\right) \geq 0.3 \\ 1 & \text{Age} \in [12, 24) \wedge \mathcal{C}[\text{Creatinine}]\left(t; m_{\text{renal}}\right) \geq 0.4 \\ 1 & \text{Age} \in [24, 60) \wedge \mathcal{C}[\text{Creatinine}]\left(t; m_{\text{renal}}\right) \geq 0.6 \\ 1 & \text{Age} \in [60, 144) \wedge \mathcal{C}[\text{Creatinine}]\left(t; m_{\text{renal}}\right) \geq 0.7 \\ 1 & \text{Age} \in [144, 216) \wedge \mathcal{C}[\text{Creatinine}]\left(t; m_{\text{renal}}\right) \geq 1.0 \\ 0 & \text{otherwise.} \end{cases} \tag{57}$$

4 Suspected Infection

A patient is defined to have a suspected infection if they have at least one infectious test *ordered* and at least one dose of a systemic anti-infective agent. Results of the tests are not needed, just that the tests have been ordered. Anti-infective agents include anti-bacterials, anti-fungals, anti-virals, or anti-mycobacterial (anti-buberculosis). This definition is operational: it identifies when clinicians treated the case as infectious, not whether infection was ultimately confirmed.

Let t_1 denote the time when an infectious test was ordered. Let t_2 denote the time when the first dose of a systemic anti-infectious agent was administered.

Suspected infection status is defined as:

$$\text{SI}\left(T\right) \coloneqq \mathbb{I}\left\{t_1 \in T\right\} \times \mathbb{I}\left\{t_2 \in T\right\}. \tag{58}$$

That is, a patient is considered to have a suspected infection if during the time interval T at least one dose of systemic anti-infective agent and at least one infectious test has been ordered.

5 Organ Dysfunction Summary Score and Phoenix Sepsis Score

This section combines the organ dysfunction scores with suspected infection to define the Phoenix Sepsis Score, sepsis, and septic shock. It also distinguishes the organ dysfunction summary score (ODSS), which can be computed regardless of suspected infection status, from the Phoenix Sepsis Score (PSS), which is gated by suspected infection.

Define the look-back windows as the vector:

$$\boldsymbol{m} \coloneqq \left(m_{\text{resp}}, m_{\text{vaso}}, m_{\text{bp}}, m_{\text{lac}}, m_{\text{gcs}}, m_{\text{pupil}}, m_{\text{coag}}, m_{\text{endo}}, m_{\text{immu}}, m_{\text{hepatic}}, m_{\text{renal}}\right). \tag{59}$$

For notational compactness, $\mathcal{D}\left[x\right]\left(t; \boldsymbol{m}, \text{Age}\right)$ uses a common argument list for all scoring entities x. Each scoring function depends only on the elements of $\boldsymbol{m}$, and on Age, that are relevant to its definition; all other arguments are ignored.

The ODSS permits organ-dysfunction burden to be summarized independently of suspected infection status. However, ODSS is introduced here as a convenience measure and has not been evaluated as a standalone clinical score; the Phoenix development work evaluated the underlying organ dysfunction score in conjunction with suspected infection [Sanchez-Pinto et al., 2024, Schlapbach et al., 2024].

$$\text{ODSS}_d\left(T; \boldsymbol{m}, \text{Age}\right) \coloneqq \max_{t \in T} \sum_{x \in \Omega_d} \mathcal{D}\left[x\right]\left(t; \boldsymbol{m}, \text{Age}\right), \quad d \in \left\{4, 8\right\}, \tag{60}$$

where

$$\Omega_4 \coloneqq \left\{\text{Resp}, \text{Card}, \text{Neuro}, \text{Coag}\right\} \tag{61}$$

and

$$\Omega_8 \coloneqq \Omega_4 \cup \left\{\text{Endo}, \text{Immu}, \text{Hepatic}, \text{Renal}\right\}. \tag{62}$$

The Phoenix Sepsis Score is then:

$$\text{PSS}_d\left(T; \boldsymbol{m}, \text{Age}\right) \coloneqq \text{SI}\left(T\right) \text{ODSS}_d\left(T; \boldsymbol{m}, \text{Age}\right), \quad d \in \left\{4, 8\right\}. \tag{63}$$

Sepsis is defined as organ-dysfunction in the presence of a suspected infection, that is:

$$\text{Sepsis}\left(T; \boldsymbol{m}, \text{Age}, \sigma\right) \coloneqq \mathbb{I}\left\{\text{PSS}_4\left(T; \boldsymbol{m}, \text{Age}\right) \geq \sigma\right\}. \tag{64}$$

Septic shock is sepsis with at least κ point(s) from cardiovascular dysfunction.

$$\begin{aligned} \text{SepticShock}\left(T; \boldsymbol{m}, \text{Age}, \sigma, \kappa\right) \coloneqq{}& \text{SI}\left(T\right) \\ & \times \mathbb{I}\left\{\max_{t \in T}\left(\sum_{x \in \Omega_4} \mathcal{D}\left[x\right]\left(t; \boldsymbol{m}, \text{Age}\right) \mathbb{I}\left\{\mathcal{D}\left[\text{Card}\right]\left(t; \boldsymbol{m}, \text{Age}\right) \geq \kappa\right\}\right) \geq \sigma\right\}, \end{aligned} \tag{65}$$

where $\sigma \geq \kappa > 0$. Note that the specific shock definition does not simply consider any cardiovascular dysfunction in the window T. It must occur with the organ dysfunction associated with reaching the sepsis score threshold.

For the published Phoenix Sepsis Criteria [Sanchez-Pinto et al., 2024, Schlapbach et al., 2024] the following parameter values in Table 11 were used. Note that the relative staleness parameters are introduced in this document to help guide prospective or real-time assessments. The default values of ∞ preserve definitions used when Phoenix was developed.

Table 11: Parameter values defining time windows, look-backs, and thresholds for the Phoenix Sepsis Score, Sepsis, and Septic Shock.

Parameter	Value	Notes
Time Window		
T	$[0, 1440)$	The first 24 hours of the encounter
Thresholds		
σ	2	
κ	1	
Look-backs		
m_{resp}	360 minutes	all respiratory EHR variables
m_{vaso}	720 minutes	vasoactive medications
m_{bp}	360 minutes	blood pressure / MAP
m_{lac}	360 minutes	lactate
m_{gcs}	720 minutes	GCS components or total
m_{pupil}	720 minutes	pupils
m_{coag}	1440 minutes	coagulation labs
m_{endo}	720 minutes	glucose
m_{immu}	1440 minutes	ANC and ALC
m_{hepatic}	1440 minutes	ALT and Total bilirubin
m_{renal}	1440 minutes	creatinine
Freshness / Relative Staleness Parameters		
$\delta_{\text{PaO}_2,\text{SpO}_2}$	∞	Introduced for prospective implementations
δ_{sdbp}	∞	Introduced for prospective implementations
δ_{MAP}	∞	Introduced for prospective implementations

Importantly, restricting the maximization to $t \in T$ does not truncate the underlying data. For any $t \in T$, the carry-forward operator $\mathcal{C}[\cdot](t; m)$ can still use measurements recorded before T (even at negative times) as long as they fall within the look-back window m. Conceptually, the ODSS corresponding to the published aggregation [Sanchez-Pinto et al., 2024, Schlapbach et al., 2024] is time-aligned: at each time t we sum the organ dysfunction components evaluated at that same time, then take the maximum of that sum over T. This preserves temporal alignment across organ systems, so a high score reflects concurrent multi-organ dysfunction at a single time point.

6 Exploratory Aggregation Schemes

There are several exploratory aggregation schemes. Daily and block-based "worst value" aggregation is common in pediatric organ dysfunction scoring (e.g., PELOD/qPELOD and pSOFA), which motivates the time-decoupled exploratory aggregation schemes below. See, for example, daily PELOD scoring using the most abnormal value each day and pSOFA evaluations that use maximum/mean daily scores or the highest score in fixed time blocks. [Leteurtre et al., 2003, Bestati et al., 2010, Matics and Sanchez-Pinto, 2017, Badke et al., 2025]

6.1 Organ-Level Maxima

The primary definition for ODSS in (60) looks for the max of the sum of the organ dysfunction scores. This exploratory scheme looks at the sum of the max of each of the organ system dysfunction scores.

$$\mathrm{ODSS}_d^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right) \coloneqq \sum_{x \in \Omega_d} \max_{t \in T} \mathcal{D}\left[x\right]\left(t; \boldsymbol{m}, \mathrm{Age}\right), \quad d \in \left\{4, 8\right\}. \tag{66}$$

$\mathrm{ODSS}_d^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right)$ decouples time across organ systems. It takes the maximum score for each organ system independently and then sums those maxima. This measures the overall burden of dysfunction over the interval, not necessarily dysfunction occurring at the same time. As a result, $\mathrm{ODSS}_d^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right)$ can be larger than $\mathrm{ODSS}_d\left(T; \boldsymbol{m}, \mathrm{Age}\right)$, because peaks from different time points can accumulate. That is, by construction,

$$\mathrm{ODSS}_d\left(T; \boldsymbol{m}, \mathrm{Age}\right) \leq \mathrm{ODSS}_d^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right). \tag{67}$$

This follows from the max-sum inequality:

$$\max_t \sum_x f_x(t) \leq \sum_x \max_t f_x(t). \tag{68}$$

The exploratory version of the PSS defined in (63) is:

$$\mathrm{PSS}_d^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right) \coloneqq \mathrm{SI}\left(T\right) \mathrm{ODSS}_d^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right), \quad d \in \left\{4, 8\right\}. \tag{69}$$

Using (69) to define sepsis is not structurally different from the primary definition (64).

$$\mathrm{Sepsis}^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}, \sigma\right) \coloneqq \mathbb{I}\left\{\mathrm{PSS}_4^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right) \geq \sigma\right\}. \tag{70}$$

Septic shock, for organ-level maxima, is an easier calculation than the primary definition (65). Since each organ system is considered independently over T then we only need to know that the contribution from the cardiovascular system is at least κ.

$$\mathrm{SepticShock}^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}, \sigma, \kappa\right) \coloneqq \mathbb{I}\left\{\max_{t \in T} \mathcal{D}\left[\mathrm{Card}\right]\left(t; \boldsymbol{m}, \mathrm{Age}\right) \geq \kappa\right\} \mathrm{Sepsis}^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}, \sigma\right), \tag{71}$$

for $\sigma \geq \kappa > 0$.

6.2 Organ-Level + Cardiovascular Component Decoupling

In this exploratory aggregation scheme we decouple all organ systems as done in the OLM and decouple the three cardiovascular components: vasoactive medications, MAP, and lactate within the cardiovascular system. Note: vasoactives can be decoupled further than they are in this exploratory aggregation scheme. In this scheme, at time t, we assign 0 points for zero medications, 1 point for one medication, and 2 points for two or more concurrent medications. In § 6.3 we will fully decouple the vasoactive medications.

$$\begin{aligned} \mathrm{ODSS}_d^{\mathrm{CCD}}\left(T; \boldsymbol{m}, \mathrm{Age}\right) \coloneqq & \sum_{x \in \Omega_d \setminus \left\{\mathrm{Card}\right\}} \max_{t \in T} \mathcal{D}\left[x\right]\left(t; \boldsymbol{m}, \mathrm{Age}\right) \\ & + \sum_{x \in \boldsymbol{C}_{\mathrm{card}}} \max_{t \in T} \mathcal{D}\left[x\right]\left(t; \boldsymbol{m}, \mathrm{Age}\right), \quad d \in \left\{4, 8\right\}. \end{aligned} \tag{72}$$

This definition leads to the relationship:

$$\mathrm{ODSS}_d\left(T; \boldsymbol{m}, \mathrm{Age}\right) \leq \mathrm{ODSS}_d^{\mathrm{OLM}}\left(T; \boldsymbol{m}, \mathrm{Age}\right) \leq \mathrm{ODSS}_d^{\mathrm{CCD}}\left(T; \boldsymbol{m}, \mathrm{Age}\right). \tag{73}$$

This exploratory ODSS leads to the exploratory PSS definition:

$$\mathrm{PSS}_d^{\mathrm{CCD}}\left(T;\boldsymbol{m},\mathrm{Age}\right) \coloneqq \mathrm{SI}\left(T\right)\mathrm{ODSS}_d^{\mathrm{CCD}}\left(T;\boldsymbol{m},\mathrm{Age}\right), \quad d \in \{4,8\}. \tag{74}$$

The definition of sepsis is structurally the same as the primary and OLM variants.

Septic shock definition for this exploratory aggregation scheme also decouples the contributions of the vasoactives, MAP, and lactate.

$$\mathrm{Sepsis}^{\mathrm{CCD}}\left(T;\boldsymbol{m},\mathrm{Age},\sigma\right) \coloneqq \mathbb{I}\left\{\mathrm{PSS}_4^{\mathrm{CCD}}\left(T;\boldsymbol{m},\mathrm{Age}\right) \geq \sigma\right\}, \tag{75}$$

and

$$\mathrm{SepticShock}^{\mathrm{CCD}}\left(T;\boldsymbol{m},\mathrm{Age},\sigma,\kappa\right) \coloneqq \mathrm{Sepsis}^{\mathrm{CCD}}\left(T;\boldsymbol{m},\mathrm{Age},\sigma\right)\mathbb{I}\left\{\sum_{x\in\boldsymbol{C}_{\mathrm{card}}}\max_{t\in T}\mathcal{D}\left[x\right]\left(t;\boldsymbol{m},\mathrm{Age}\right) \geq \kappa\right\}, \tag{76}$$

for $\sigma \geq \kappa > 0$.

6.3 Full Component Decoupling

Under full component decoupling, each point-producing component is aggregated independently over T and then combined using the original organ-specific capping or maximum rules. Component scores are maximized over T; for respiratory oxygenation, lower PFR and SFR values represent greater dysfunction and are therefore minimized. This scheme decouples respiratory support from oxygenation severity; the required support and oxygenation threshold need not occur at the same time within T.

$$\begin{aligned}\mathrm{Resp}^{\mathrm{FCD}}\left(T;m_{\mathrm{resp}}\right) \coloneqq\ & \max_{t\in T}\left(\mathrm{IMV}\left(t;m_{\mathrm{resp}}\right)\right) \times \left(\mathbb{I}\{B_3(T;m_{\mathrm{resp}})\} + \mathbb{I}\{B_2(T;m_{\mathrm{resp}})\}\right) \\ & + \max_{t\in T}\left(\mathrm{ORS}\left(t;m_{\mathrm{resp}}\right)\right) \times \mathbb{I}\{B_1(T;m_{\mathrm{resp}})\},\end{aligned} \tag{77}$$

where

$$\begin{aligned}B_1(T;m_{\mathrm{resp}}) &\coloneqq \min_{t\in T}\mathrm{PFR}\left(t;m_{\mathrm{resp}}\right) < 400 \vee \min_{t\in T}\mathrm{SFR}\left(t;m_{\mathrm{resp}}\right) < 292, \\ B_2(T;m_{\mathrm{resp}}) &\coloneqq \min_{t\in T}\mathrm{PFR}\left(t;m_{\mathrm{resp}}\right) < 200 \vee \min_{t\in T}\mathrm{SFR}\left(t;m_{\mathrm{resp}}\right) < 220, \\ B_3(T;m_{\mathrm{resp}}) &\coloneqq \min_{t\in T}\mathrm{PFR}\left(t;m_{\mathrm{resp}}\right) < 100 \vee \min_{t\in T}\mathrm{SFR}\left(t;m_{\mathrm{resp}}\right) < 148.\end{aligned} \tag{78}$$

For vasoactive medications, each individual medication indicator is maximized independently resulting in:

$$\mathrm{Vasos}^{\mathrm{FCD}}(T;m_{\mathrm{vaso}}) \coloneqq \min\left\{2, \sum_{v\in\boldsymbol{V}}\max_{t\in T}\mathbb{I}\left\{\mathcal{C}[v](t;m_{\mathrm{vaso}}) = 1\right\}\right\}. \tag{79}$$

The fully decoupled ODSS is:

$$\begin{aligned}\mathrm{ODSS}_d^{\mathrm{FCD}}\left(T;\boldsymbol{m},\mathrm{Age}\right) \coloneqq\ & \mathrm{Resp}^{\mathrm{FCD}}\left(T;m_{\mathrm{resp}}\right) + \mathrm{Vasos}^{\mathrm{FCD}}(T;m_{\mathrm{vaso}}) \\ & + \min\left\{2, \sum_{x\in\boldsymbol{C}_{\mathrm{coag}}}\max_{t\in T}\mathcal{D}\left[x\right]\left(t;m_{\mathrm{coag}}\right)\right\} \\ & + \sum_{x\in\omega_d}\max_{t\in T}\mathcal{D}\left[x\right]\left(t;\boldsymbol{m},\mathrm{Age}\right), \quad d \in \{4,8\}.\end{aligned} \tag{80}$$

where

$$\omega_4 \coloneqq \{\mathrm{MAP}, \mathrm{Lactate}, \mathrm{Neuro}\}, \tag{81}$$

and

$$\omega_8 \coloneqq \omega_4 \cup \{\text{Endo}, \text{Immu}, \text{Hepatic}, \text{Renal}\}\,. \tag{82}$$

Note: the contributions from neurologic, immunologic, and hepatic scores do consider more than one input from the EHR. However, because

$$\begin{aligned}\max_{t\in T}\max\left\{\mathcal{D}\left[\text{GCS}\right](t;m_{\text{gcs}}), \mathcal{D}\left[\text{Pupils}\right](t;m_{\text{pupil}})\right\} = \max\Bigg\{&\max_{t\in T}\mathcal{D}\left[\text{GCS}\right](t;m_{\text{gcs}}),\\ &\max_{t\in T}\mathcal{D}\left[\text{Pupils}\right](t;m_{\text{pupil}})\Bigg\}.\end{aligned} \tag{83}$$

we need not explicitly decouple these component scores.

The construction of this exploratory aggregation scheme creates the relationship:

$$\text{ODSS}_d\left(T;\boldsymbol{m},\text{Age}\right) \le \text{ODSS}_d^{\text{OLM}}\left(T;\boldsymbol{m},\text{Age}\right) \le \text{ODSS}_d^{\text{CCD}}\left(T;\boldsymbol{m},\text{Age}\right) \le \text{ODSS}_d^{\text{FCD}}\left(T;\boldsymbol{m},\text{Age}\right). \tag{84}$$

The exploratory Phoenix sepsis score is:

$$\text{PSS}_d^{\text{FCD}}\left(T;\boldsymbol{m},\text{Age}\right) \coloneqq \text{SI}\left(T\right)\text{ODSS}_d^{\text{FCD}}\left(T;\boldsymbol{m},\text{Age}\right), \quad d\in\{4,8\}\,. \tag{85}$$

and Sepsis is defined as:

$$\text{Sepsis}^{\text{FCD}}\left(T;\boldsymbol{m},\text{Age},\sigma\right) \coloneqq \mathbb{I}\left\{\text{PSS}_4^{\text{FCD}}\left(T;\boldsymbol{m},\text{Age}\right)\ge\sigma\right\}, \tag{86}$$

for $\sigma > 0$.

Septic shock is defined as follows:

$$\begin{aligned}\text{SepticShock}^{\text{FCD}}\left(T;\boldsymbol{m},\text{Age},\sigma,\kappa\right) \coloneqq{}& \text{Sepsis}^{\text{FCD}}\left(T;\boldsymbol{m},\text{Age},\sigma\right)\\ &\times\mathbb{I}\left\{\text{Vasos}^{\text{FCD}}(T;m_{\text{vaso}}) + \sum_{x\in\{\text{MAP},\text{Lactate}\}}\max_{t\in T}\mathcal{D}\left[x\right](t;\boldsymbol{m},\text{Age})\ge\kappa\right\}\end{aligned} \tag{87}$$

for $\sigma \ge \kappa > 0$.

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